

SYSTEM ID REPORT

(ROCKET LANDING SYSTEM)

1. SYSTEM DESCRIPTION

The system is a **rocket landing model** designed to simulate vertical take-off, hover, and controlled landing using thrust vector control.

The rocket is modeled as a **rigid body system** and is controlled using **PID-based feedback control** implemented in a Python–PyBullet environment.

The system accounts for **gravity, drag, and wind disturbances**, enabling realistic simulation of aerospace landing dynamics.

The primary objective is to achieve **stable hover and precise landing** through control of thrust and orientation.

2. INPUT VARIABLES (I)

Control Inputs:

- Thrust $T \rightarrow$ controls vertical motion
- Gimbal angle $\delta \rightarrow$ controls tilt and direction

Disturbances:

- Wind force (external disturbance)
- Aerodynamic drag

Control Parameters:

- $K_p \rightarrow$ proportional gain
 - $K_d \rightarrow$ damping gain
-

3. OUTPUT VARIABLES (O)

Motion Outputs:

- Position $\rightarrow x(t), z(t)$
- Velocity $\rightarrow \dot{x}, \dot{z}$
- Orientation $\rightarrow \theta(t)$

Performance Outputs:

- Hover stability
 - Landing accuracy
 - Final velocity at touchdown
-

4. ASSUMPTIONS

- Rocket behaves as a rigid body
 - Motion is planar (no yaw)
 - Small angle approximation is valid
 - Thrust acts along rocket body axis
 - Drag is proportional to velocity
 - Mass remains constant
 - External disturbances modeled as wind forces
-

5. FUNCTIONAL BLOCK DIAGRAM (FBD)

INPUTS (Controller + Environment)

→ Thrust T , Gimbal δ , Wind



CONTROLLER (PID CONTROL)

→ Uses K_p, K_d to stabilize tilt

→ Controls altitude and descent



ROCKET DYNAMICS (PLANT)

→ Translational motion (x, z)

→ Rotational motion (θ)

→ Includes gravity and drag



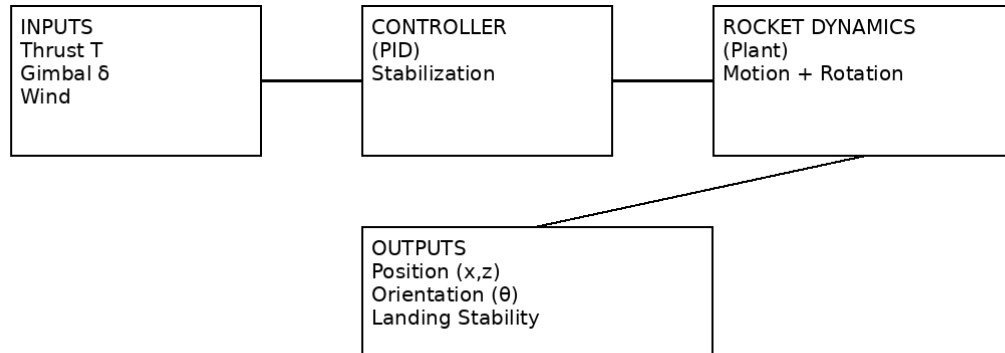
OUTPUTS (SYSTEM RESPONSE)

→ Position x, z

→ Orientation θ

→ Stable landing

Functional Block Diagram (FBD)



6. WORKING PRINCIPLE

The system operates using **closed-loop feedback control**.

- During lift-off, thrust overcomes gravity
- During hover, thrust balances forces
- During descent, thrust is reduced to control velocity

The controller continuously adjusts thrust and gimbal angle based on system state to ensure **stable and accurate landing despite disturbances**.

MADE BY:

Siddharth Bathija (H004)

Jayhan Bhagvati (H005)

Jimit Kacha (H028)

Shehzad Khan (H033)